DOST Form 3

NON-R&D PROJECT PROPOSAL

(Technology Transfer, S&T Promotion and Linkages, Policy Advocacy, Provision of S&T Services, Human Resource Development and Capacity-Building)

I. PROJECT PROFILE

(1) Program Title: Regional Agri-Aqua Innovation System Enhancement (RAISE)
Project Title: Regional Innovation and Technology Center (RITC)

(2) Project Leader/Sex: Prof. Michael C. Barbecho
Agency: Partido State University.
Address/Telephone/Fax/Email: Goa Camarines Sur

(3) Cooperating Agency: BU, CNSC, CBSUA, CSPC, DEBESMSCAT, CatSU, SorSU

(4) Implementing Agency: Partido State University
Address/Telephone/Fax/Email: Goa, Camarines Sur
Base Station: Goa, Camarines Sur
Other Implementation Site (s): _____

(5) Project Duration (number of months): _____
Project Start Date: January 1, 2022
Project End Date: December 31, 2023

(6) Total Project Cost: _____ (indicate Counterpart Funds; use Form 4 for the Line-Item Budget)

Implementing Agency/ies	PS	MOOE	EO	Total
A. Requested Fund				
DOST-PCAARRD	755,097.60	834,902.40	210,000.00	1,800,000.00
B. Counterpart Fund 1				
ParSU		480,000.00	-	480,000.00
TOTAL	**755,097.60**	**1,314902.40**	**210,000.00**	**2,280,000.00**

Sustainable Development Goal (SDG) Addressed
_____

II. PROJECT SUMMARY

(7) Executive Summary:

Patido State University (ParSU) integrated the Research and Development function since 1994. This integral component to be functional must conduct researches, projects and programs responsive and directed towards enhancement of the existing programs of the university and supportive to the country's development goals. Research management at ParSU must enunciate clearly its vision, mission goals and objectives, areas of concern, thrusts and priorities, organizational structure and functions, and implementing mechanisms. This is done to ensure an orderly manner and professional conduct of research in this academic community.

ParSU evolved and implemented an effective mechanism for forecasting, planning, policy making, financing managing, and assessing Research Activities designed to promote the institutions growth and development along the four goals of higher education: access and equity, quality and excellence, relevance and responsiveness, and efficiency and effectiveness, and along the priority areas of the local government and national government. It also want to gain recognition as a reputable Research unit with a track record of conducting Research activities that will contribute to the pool of knowledge, and solve the problem of the academe, community

ParsU is also conduct research and development studies in social sciences, business, science and technology, culture and arts, agriculture and fisheries. It will strengthen the manpower and resource capabilities of the university. It will formulate and recommend research policies and guidelines conducive to and supportive of the research endeavor of the faculty and other members of the academe. It will establish linkage with government and non-government agencies, industry and business sectors, sister SUC's and other local and foreign institution for fund sourcing, capability building, and networking along the research and development endeavor. ParSU will also disseminate information and technologies responsive to the emerging needs of its clientele. It will also create mechanisms for monitoring and evaluating researches and research activities, and for dissemination, implementation and utilization of research outputs.

With the aforementioned vision, ParSU sees the need seminars and trainings on IP protection and Commercialization that will address the lacking avenues for technology partnerships and commercialization of AANR products of the institution. The RAISE Program can provide the capability training and building on technology transfer to R&D partners while developing ParSU's establishment of technology transfer office. This will also bring forth a maximized economic potential of science and technology innovations.

(8) Introduction

Rationale/Significance (Not to exceed 300 words)

The Philippines has further increased its innovation achiever position in the 2020 Global Innovation Index (GII) posting a jump from the 50th place from the 54th in the previous year and is considered the highest position ever of the country. The GII ranks 131 world economies based on innovative capabilities. The innovation inputs and outputs are being considered in the GII.

This achievement has been achieved due to the implementation of various laws that strengthened the scope of protection and support to IP protection and commercialization. One of these laws include the RA 10055: Technology Transfer Act which promote the utilization of intellectual property for the national benefit and shall call upon all research and development institutes and/or institutions (RDIs) that perform government-funded research and development (R&D) to take on technology transfer as their strategic mission and to effectively translate results of government-funded R&D into useful products and services that will redound to the benefit of Filipinos, notwithstanding the income generated from intellectual property rights (IPRs) and technology transfer activities. RA 10372: Revised IP Code further defines and strengthen the IP Code of the Philippines and the Intellectual Property Office of the Philippines to facilitate the protection of products and technologies produced in the Philippines. RA 11293: Innovation Act on the other hand aims to foster innovation as a vital component of national development and sustainable economic growth, while RA 11337: Innovative Start-up Act is to foster inclusive growth through an innovative economy by streamlining government and nongovernment initiatives, in both local and international spheres, to create new jobs and opportunities, improve production, and advance innovation and trade in the country.

DOST as one of the identified "start-up enablers" in the country has made significant contribution to this achievement though various initiatives. In 2016, PCAARRD launched the

DOST-PCAARRD Innovation and Technology Center (DPITC) that caters to the technology commercialization initiatives of the agency. Under the initiatives of the DPITC, the Intellectual Property-Technology Business Management (IP-TBM) program was conceptualized and approved. For the batch 1 of the program 5 licensing agreements were executed against the target of 10. for Batch 2, exploratory meetings with potential investors are still on-going. To sustain and pursue the initial efforts in protecting the technologies generated by the participating HEIs in the Batch 1 and 2 of the IP-TBM program. The SUSTAIN IP-TBM Phase II was implemented using the mentor-mentee approach where there are five mentor agencies that taught and guided thirty mentor agencies on IP Protection and Commercialization.

In Bicol region, Patido State University (ParSU) integrated the Research and Development function since 1994. This integral component to be functional must conduct researches, projects and programs responsive and directed towards enhancement of the existing programs of the university and supportive to the country's development goals. Research management at ParSU must enunciate clearly its vision, mission goals and objectives, areas of concern, thrusts and priorities, organizational structure and functions, and implementing mechanisms. This is done to ensure an orderly manner and professional conduct of research in this academic community.

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With the aforementioned vision, ParSU sees the need seminars and trainings on IP protection and Commercialization that will address the lacking avenues for technology partnerships and commercialization of AANR products of the institution. The RAISE Program can provide the capability training and building on technology transfer to R&D partners while developing ParSU’s establishment of technology transfer office. This will also bring forth a maximized economic potential of science and technology innovations.

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Objectives:

1. Facilitate public-private access to AANR technologies to improve the innovation ecosystem in the region;
2. Provide a venue for convergence of regional AANR stakeholders from the academe, public, private sectors, NGOs and international partners;
3. Manage regional AANR technologies and Intellectual Properties;
4. Provide capability building on tech transfer to R&D partners;
5. Strengthen existing and forge new Public Private Partnerships for RDRU;
6. Support regional partners in developing their respective tech transfer programs

Methodology:

1. Capacity Building and Mentorship
2. Agri technology Business Development
3. Intellectual Property Management
4. Strategic partnership and Collaboration
5. Enhance Knowledge Management

Expected Outputs (6Ps):

Expected Output Project 1 (Old CMI)		
Y1 Output	**Y2 Output**	**Total**
Publications		
1 Training Module on IP Management 3 IEC	 2 IEC 1 Regional Sustainability Plan	1 Training Module on IP Management 5 IEC 1 Regional Sustainability Plan
Patents/ IP		
5 IP Applications	5 IP Applications	10 IP Applications
Product		
5 Prior art search of R&D Proposals 1 Institutional inventory of potential IPs 1 Institutional Inventory of IP Assets	5 Prior art search of R&D Proposals 1 Institutional inventory of potential IPs 1 Institutional Inventory of IP Assets	10 Prior art search of R&D Proposals 2 Institutional inventory of potential IPs 2 Institutional Inventory of IP Assets

1 Institutional Inventory of Mature Technologies 1 Product manufactured for pre-comm/ market tested 1 technology with pre-comm reports 1 Technology Commercialized	1 Institutional Inventory of Mature Technologies 1 Product manufactured for pre-comm/ market tested 1 technology with pre-comm reports 1 Technology Commercialized	2 Institutional Inventory of Mature Technologies 2 technology with pre-comm reports 2 Product manufactured for pre-comm/ market tested 2 Technology Commercialized
People and Services		
1 Institutional Prior Art Search & IP Audit Send at least 2 CMI Staff to IP Masterclass	At least 2 CMI Staff send to be trained for Technology Commercialization Mentorship Series 1 Institutional Pitch Day	1 Institutional Prior Art Search & IP Audit Send at least 2 CMI Staff to IP Masterclass At least 2 CMI Staff send to be trained for Technology Commercialization Mentorship Series 1 Institutional Pitch Day
Places and Partnerships		
1 Commercialization Agreement Signed	1 partnership agreement with Business Groups/Trade Institutions	1 Commercialization Agreement Signed 1 partnership agreement with Business Groups/Trade Institutions
Policy		

Potential Outcomes:

1. Increased number of IP applications and registered IP for the institution's agri-aqua technologies;
2. Integrated DOST-PCAARRD Innovation and Technology Center (DPITC) functions in the IP-TTO of the university.
3. Trained staff that will be in charge of DPITC-based functions;
4. Partnerships and agreements for commercialization with technology adaptors;

Potential Impacts (2Is):

1. Increase number of AANR technologies developed and protected in the region.
2. Hasten the technology/IP commercialization and building up S&T Based Agri-Aqua Enterprises of the AANR Sector in the Region
3. Proliferation of knowledge on innovation ecosystem and maximize economic potential of

S&T Innovation

Discussion on the results of related project handled by the same proponent (if any):

1. Partido State University (ParSU) sees the importance to become one of the mentee of the RAISE program to avail the needed seminars and trainings on IP protection and Commercialization that will address the lacking avenues for technology partnerships and commercialization of AANR products of the institution. The RAISE Program can provide the capability training and building on technology transfer to R&D partners while developing ParSU's establishment of technology transfer office. This will also bring forth a maximized economic potential of science and technology innovations.

Target Beneficiaries:

The RAISE program will benefit the institution especially faculty members researchers, and students who will pursue research to create technological products. The IP application and technology commercialization will safeguard the research products of the university while earning revenue. This will also benefit the community who will be part of the production team and the target community for the use of the technological product.

The completion of the program and the continuous efforts to bringing the knowledge and technologies closer to the industries through technology business incubation in the realization of their goals to be a conduit of connection and collaboration between the mentor and mentee agencies, Department of Agriculture Region V, Department of Science and Technology Region V, Department of Trade in Industry Region V, City LGU and Municipal LGU in the six Provinces of Bicol Region, private sector and MSMEs and the incubatees.

The industries, private sectors and MSMEs to be able to address their needs in increasing their efficiency and effectiveness without compromising quality over quantity and productivity and building valuable partnerships with them.

The Faculty and students by making them researches realize its societal impact anchored in the mind to market concept. Plus, be able to combat the status quo that is the "Publish and Perish" mentality through technopreneurship and technology business incubation.

The Alumni by strengthening our ties and recognizing their roles as key partners in establishing the university incubator and building a solid platform for helping ideas shape into commercial ventures through the right kind of support system and mentorship in addition to commercializing research.

Partido State University in achieving its vision of establishing an AANR start-

up/spin- off company through technology business incubation and to institutionalize a technopreneurship curriculum that will be thought in the program. Also, the research output of both faculty and students that will be used by the industries and private, and individuals in a day- to-day basis is a desired impact of following the mind to market concept of the program.

This is in support with the performance targets and commitment of the different units of Partido State University towards SUC levelling since commercialization of technologies is one of the key metrics measured in SUC Leveling.

Finally, this is for the thriving incubation sector and community in the Philippines. In order for them to realize their full potential through the pool of mentors, network of partners and investors and the incubation program Bicol University shall offer. Because at ParSU, we believe that start-ups are the way to go.

Sustainability Plan:

The establishment of the Regional Agri-Aqua Innovation System Enhancement (RAISE) Bicol will help the researchers, inventors and start-up technology-based businesses to provide a range of resources, services, and facilities needed from development to commercialization stage of agriculture and aquaculture technologies. With Partido State University as one of the agri-aqua technologies research hubs in the Bicol Region, the institution creates and innovates a variety of technologies and innovations that benefit the community and the country.

The project will be relying on the government support fund. Later on, the office will be an attached office to the BCAARRD wherein its funding support will rely according to its accomplishment. Yearly, as a requirement for graduating students of Partido State University is to produce and generate inventions and innovations of agri-aqua technologies that will bring huge impact to the society. the created prototypes and outputs of the students will be audited to be part of the Intellectual Property (IP) inventories of the school and will be further evaluated if qualified for technology protection and commercialization. Through the Technopreneurship and in partnership with graduating agribusiness students, the protected IP's will be commercialized by establishing start-up companies that will ensure that the technology will be utilized by the prospected end users. Thus, the program will have an income from the percent in licensing as its operational fund. Also, the TBI operation will be able to continue to support and operate the IP generation and management of the technologies of the university for a longer period of time. According to the IPOPhil and ParSU IP Policy which is also the basis of IP Policy approved in each CMI,

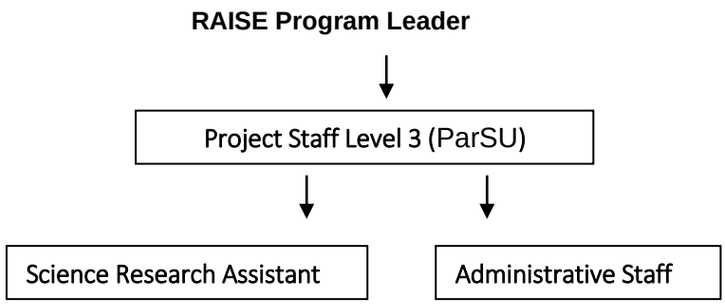
there will be payment for IP professional/Patent agent that will handle IP Protection activities. Also, ParSU aim to establish ATBI that will serve as its commercialization hub shall collect payment after the 3rd year of operation to its client. Therefore, percentage of that fund will support in the operation of the propose project.

Gender and Development (GAD) Score (refer to the attached GAD checklist):

Literature Cited:

(9) Workplan (See Form 5)

(10) Project Management (not to exceed one page)



Duties and Responsibilities:
Project Staff Level 3: oversees the course of the project, and reports the progress of the project to the program leader to ensure that all objectives are attained and all outputs are accomplished.
Science Research Assistant: assists in the preparation of communication and financial and progress reports and supports the operation of the project component.
Administrative Staff: serves as the counterpart of the university in the project

(7) Rationale

The Philippines marked its first innovation achiever position in the 2020 Global Innovation Index, posting a big jump to 50th place from the previous year's 54th position as it catches up with world leaders (www.pna.gov.ph). With a total score of 35.19 over 131, the report said the Philippines is among the countries that have "above expectations for level of development" among lower-middle-income economies.

The Philippines ranked 13th among southeast Asian and Oceania countries and 70th in the world on the innovation input sub-index after getting a score of 40.75 points, which, however, remains below the median score of 43.46. In terms of the innovation output sub-index, the country ranked 42nd in the world and 10th in the region after getting a score of 30.68, higher than the 23.54 median score. Among those belonging to the lower-middle-income economies, the Philippines landed on the 6th position for the 10 best-ranked economies on the GII, behind Vietnam, Ukraine, Georgia, India, and Mongolia.

The report said governments around the globe had increased the use of intellectual property in their quest for innovation, with investments on R&D growing more than double between 1996 and 2016. It said R&D expenditures of governments around the world rose by 5 percent while business R&D expenditures went up by 6.7 percent, the most significant jump since 2011.

DOST as one of the identified “start-up enablers” in the country has made significant contribution to this achievement though various initiatives. In 2016, PCAARRD launched the DOST-PCAARRD Innovation and Technology Center (DPITC) that caters to the technology commercialization initiatives of the agency. Under the initiatives of the DPITC, the Intellectual Property-Technology Business Management (IP-TBM) program was conceptualized and approved. For the batch 1 of the program 5 licensing agreements were executed against the target of 10. for Batch 2, exploratory meetings with potential investors are still on-going. To sustain and pursue the initial efforts in protecting the technologies generated by the participating HEIs in the Batch 1 and 2 of the IP-TBM program. The SUSTAIN IP-TBM Phase II was implemented using the mentor-mentee approach where there are five mentor agencies that taught and guided thirty mentor agencies on IP Protection and Commercialization.

In Bicol region, Bicol University has been identified as the mentor agency for the SUSTAIN IP-TBM program due to its experience and programs IP protection and commercialization. Partido State University is one of its seven (7) SUC mentee all member of Bicol Consortium for Agriculture, Aquaculture and Natural Resources Research and Development. As per IP audit of the mentor and mentee agencies, there are numerous technologies developed through the researches and studies conducted by the faculty and students of participating SUC’s in the region. While the BU as the mentor agency is now starting to commercialize its Protected IP’s the mentee SUC’s are still mastering the process. By this there is a need to continue the mentoring process and should be elevated and added agribusiness process, technology incubation process and the knowledge management process to ensure that the technology develop by faculty funded by the government will surely benefit the community.

Thus Regional Agri-Aqua Innovation System Enhancement (RAISE) Bicol program is proposed. The RAISE Bicol program will not only cater on the IP protection and pre-commercialization process of the technology, but it will cater from creating agribusiness start-up, technology incubation and knowledge management. Thus, the program will increase the number of mentee agency, increase number of IP protected, enhance the awareness in establishing start-ups, introduce technology incubation and marketing of products and technologies, and know the process and importance of knowledge management. This will zero the gap between the academe, researchers, market end-users and industry. The proposed program will develop partnership between participating agencies in the region. Through the program, the developers of the identified technologies will be assisted in the process of filing IP protection for their technology, commercializing the technology though licensing and spin-off or tart-up establishment, packaging and labeling their product, negotiating deals to champions the interests of the inventor, the agency and the client user, and the creation of inventory and technology management system.

8.1 Objectives:

1. Facilitate public-private access to AANR technologies to improve the innovation ecosystem in the region;
2. Provide a venue for convergence of regional AANR stakeholders from the academe, public, private sectors, NGOs and international partners;
3. Manage regional AANR technologies and Intellectual Properties;
4. Provide capability building on tech transfer to R&D partners;
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8.2 The RAISE Methodology:

8.2.1. Regional Capacity Building and Mentorship

Training and Capacity Building on IP Protection and Commercialization

The RAISE program has four (4) project wherein Project 1 is the Regionalization of Intellectual Property – Technology Business Management (IP-TBM), Project 2 is the Regional Agribusiness Hub, Project 3 is the Regional Agri-Aqua Technology Business Incubation, Project 4 is the Regional Knowledge Management. These four (4) projects will be based in mentor agency.

The Partido State University will select faculty and students that will be capacitated on IP management and business development/technology commercialization through the conduct of the condensed 10-module training workshop (Figure 3). The university will sent faculty and students to be mentored in technology transfer which will consist of 2 sets of training – 1) the Intellectual Property Master Class, and 2) the Technology Commercialization Mentorship Series. Both sets of training will consist of five (5) modules that will be conducted for 15 days spread over 6 months (3 days/month). The IP Master Class will demonstrate the whole patent application flow through hands-on prior art searches, actual claim drafting, filing application, substantive examination request, responding to Formality Examination Reports (FER) up to the administration of a mock exam for patent agent aspirants. The Technology Commercialization Mentorship Series will equip the trainees with the different pre-commercialization business processes, technology pitching, negotiation, and commercialization strategies. The trainees are expected to file patent or utility model applications during the training, while the prosecution of these applications will be done through mentorship by the trainers.

A training provider that will match the expertise needed will be approached to handle the IP Management training through a government-to-government contract of service. If this is not workable, public bidding will be conducted. For the business development/technology commercialization training, qualified training providers/a pool of experts will be tapped to perform the mentorship series. Contracting will be through public bidding. The series of trainings will be conducted at the identified venue in Bicol.

The project 1 team of RAISE program are composed of the personnel exposed to IP/Technology Commercialization training/conferences in coordination with PCAARRD and IPOPHL. The project will look for possible training providers where IP management and technology commercialization activities are well entrenched and practiced. Below is the training module for the IP protection and Commercialization Mentorship Series.



Figure 1. The 10-module Training Workshop on Intellectual Property Management and Technology Business Management that will adopted by RAISE program.

Partido State University will also conduct the same mentorship to its faculty and students as the participants of the mentorship series will serve as its resource speakers.

To measure the effectiveness of the training/mentorship workshops, each participating institution will be assessed according to the following indicators, but not limited to:

- Number of IP filings
- Number of echo seminars conducted as a resource person
- Number of trainees graduated

The program team shall develop a merit system among the graduates taking into consideration the commitment of the participants for the continued and real practice as patent agents.

Training and Capacity Building on Agribusiness Ventures

Partido State University will participate in the training and capacity building in the creation of the Agribusiness Start-up Companies. This will capacitate faculty participants in the establishment and operation of an agribusiness. The training will be done simultaneously with

the IP master Class wherein the target participants are those mentee and mentor who have completed the IP Master Class and IP commercialization mentorship Series. Below is the proposed module of the Training and Capacity Building for the Establishment and Operation of Start-up Company.

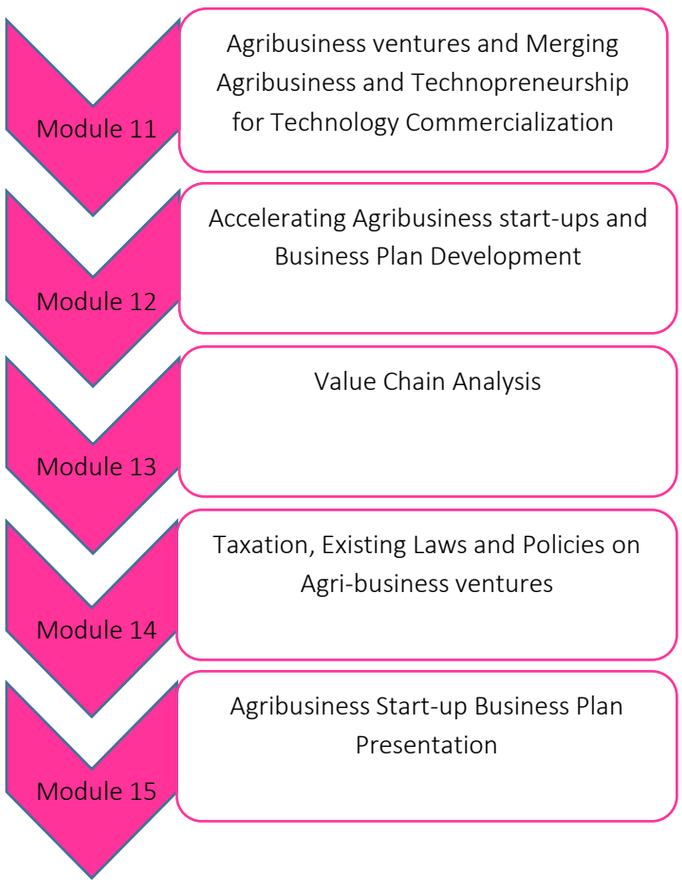


Figure 2. The propose 5-module Training Workshop on Agribusiness Ventures

8.2.2. Agri technology Business Development

Identify New Agricultural Technology that has a Viable Product for Mass Production and Marketing.

Partido State University will identify of at least two (2) agricultural technologies that has a viable product that can be commercialized and marketed in both local, national and international market. The inventors or makers of the technology will be mentored in IP protection and commercialization the technology through the 10-module Training Workshop on Intellectual Property Management and Technology Business Management of PCAARRD that will be conducted by the mentee agency. The inventors will be partnered with a graduating BS Entrepreneurship Student to help them conduct the feasibility study and make a business model for the product.

Establishment of Start-up Company from Agri-Aqua Technologies

The RAISE program will assist and train technology developers of Partido State University on valuing their developed technology through a series of trainings on IP valuation that will include the knowledge on valuation, conducting market study and generating market data. They will also assist and train technology developers to identify their users, analyzing the costumer journey, costumers empathy mapping, naming the company and organizing their business team, creating a company logo and tag line, creating a company vision, mission and goal, securing license to operate, analyzing and creating value proposition, technology valuation analysis and computation, technology competitive analysis, business model and the use of business model canvass, go-to-market plan and market share analysis, creating business plan as well as assist and train technology developer in analyzing value chain of their raw materials and product.

The RAISE program also will assist and train technology developers on crafting term sheets and licensing agreements for technology commercialization. Assist them in Technology Readiness Level Assessment, getting the Fairness Opinion Report and Attract start-up companies offering agri-aqua technologies to undergo incubation.

Conduct of Product Display, Advertisement and Marketing.

The RAISE program will identify a building and site in Partido State University that will be the display center of the technologies developed by the mentor and mentee agency. It will also urge agri-aqua technology inventors and makers to produce their product and display it to the Agribusiness hub located in the mentor and mentee agencies. It will also urge the agri-aqua inventors and makers to develop an advertising video to be uploaded to the social medias and to be played in the legitimate media outfits. It will further urge preliminary marketing of the product for market testing.

The Incubation of the Selected Technologies

The RAISE project will focus on maximizing and advancing the potential technologies and innovations generated by the faculty, researchers and students of mentor and mentee as well as Partido State University through technology transfer and commercialization by spin-off, licensing, or start-ups. The program will tap and form a pool of technology commercialization

advocates, industry collaborators, seed funding partners, and technology and business experts to be the lead promoters in diffusing technology and innovation to the market.

Technology Shopping

The Technology Portfolios will be selected from the IP asset inventory will be identified for incubation. These technologies are from agri-aqua production, agri-aqua equipment/device, agri-aqua food and feed meal, and postharvest technologies.

Business Modelling

A business plan will be developed to the identified technologies for incubation. The Business Model Canvas will be used a device to make sure that the technology or product will surely reach market and the end user and to make it sustainable. The Business Team of the incubatees will be formed and trained on product and service management, promotions, and knowledge management.

Application

Sorsogon State Univesity will endorsed a start-up that will become an incubatee of the Regional ATBI located in the mentor agency. The prospect incubatee will submit an expression of interest in a form of letter will be submitted to the Partido State University and endorsed to Bicol Agri-Aqua Technology Business Incubation office. It will be endorsement to the mentor agency, Dean, Director of Technology Commercialization, Vice President for Research, Development and Extension and to the University President for Approval. After the approval from the President the prospective incubatee will be required to submit their *Business Model Canvas* and *Team Formation. A Memorandum of Agreement will be signed together with the* Non-Disclosure Agreement. This incubatee will become a co-incubatee of the Partido State University and the mentor agency once the university is capable of handling the project.

Incubation

The Partido State University will be endorsed at least two start-up companies to become a experimental incubatee for ParSU and an outwall co-incubatee of the Bicol Agri-Aqua Technology Business Incubation (BATBI). The RAISE program and BATBI will assist in finding partners and collaborators until such time that will be able to execute a commercialization agreement with a business entity or an institutional agreement with an HEI or an RDI. The

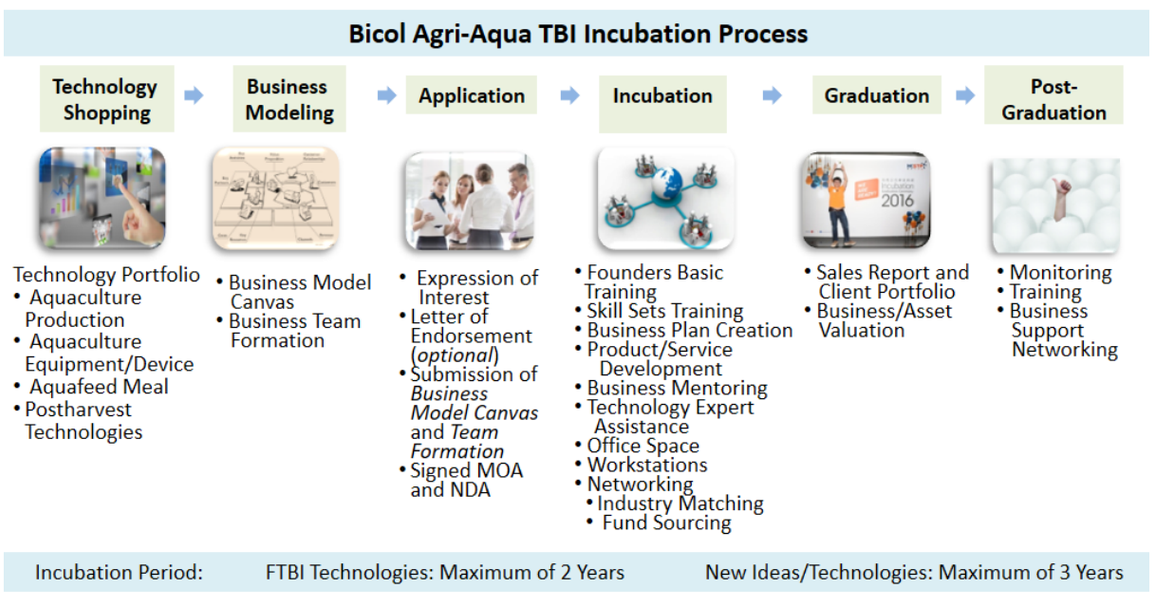
incubatees of the ParSU and BATBI will be thought with Founders Basic Training, Skill Sets Training, Business Plan Creation and Product/Service Development. Business Mentoring will also be done from time to time. Technology expert assistance, office space and workstations will be provided to incubatee. Networking and Industry Matching will also be done from time to time. Fund Sourcing will also be explored. Partido State University will be learning the process and activities of handling ATBI so that it can establish its own ATBI in the coming years.

Graduation

An incubatee will be classified as graduate they had reach the first million income. A Sales Report, Client Portfolio and Business/Asset Valuation will be required to be submitted every year to know if an incubatee will be classified as graduate.

Post-Graduation

The support to the incubatee who graduated from incubation will not end after graduation. The BATBI will still monitoring, give training and business support networking to the graduated incubatee for them to become more sustaining company.



8.2.3. Intellectual Property Management

The RAISE program will be dealing with IP Management that will facilitate the process of technology protection. Every year the project will conduct IP Audit to identify and list the technologies that need to be protected. The technologies identified for protection will be

subjected to patent search, patent drafting, patent application and prosecution. Ten (10) protected IP's will be the target of the mentor and mentee agencies for 2 years The IP's applied for protection will subjected for commercialization.

IP Disclosure

Researchers and inventors of the CMI's will be requested to submit disclosure of the agri-aqua technologies to the IP-TBM Team of the ParSU especially those technologies generated by through research funded by DOST and DOST PCAARRD. The disclosures will be the basis for assessing the output technology with its novelty, obviousness, and industrial application.

Patent Search

The disclosures will be subjected to patent search. The RAISE program will subscribe to a patent search database to make the patent search report more reliable. The patent search report will be used as basis for drafting the patent of the technology disclosed.

Patent Drafting

The patent will be drafted by the IP TBM team of the Sosogon State University. It will be checked by the Project leader before submitting to IPOPhil. The project leader will endorse its application for IP protection.

IP Application

The RAISE program will fund the IP application of agri-aqua technologies in IPOPHIL. The IP TBM staff will be the in-charge of its prosecution.

Commercialization of IP's Generated by Project Funded by PCAARRD

The Intellectual Property Commercialization will be done by establishing a start-up and incubation through the assistance of the mentor agency. Inventors will be partnered with graduating students from agribusiness or entrepreneurship to put up a company top commercialized the generated IP.

IP valuation and Socio-Economic Analysis

The project 2 will tap the SERDAC project to come up with IP valuation and Socio-Economic analysis and computation of the IP to be commercialized.

8.2.4. Strategic partnership and Collaboration

Manage Business Network/Partnerships

Exploratory meetings with potential business partners, such as but not limited to business groups, marketing, and trader's groups will be conducted. Discussions will be made with the local Chamber of Commerce, Inc. (CCI) organization to booster the commercialization of the technologies developed by mentor and mentee agency.

The RAISE program will be targeting the private sector as potential takers of the technologies that they will promote, this partnership will be fully explored to fast-track commercialization of PCAARRD-supported technologies in the SUCs/RDIs through the IP-TBM. Linkages and partnership with the industries, private sector, and other stakeholders. Collaboration with the private sectors, industries local government units, and associations will be made as partners and or prospective adaptors of the commerciable technologies generated by the participating agencies. Inter-personal visits and dialogue with the target partners and potential adopters will be conducted.

To promote the technologies developed by the SUCs/RDIs, networking cum product promotion activities will be conducted or participated by each SUC/RDI. Attendance to the NSTW, PCAARRD SiPAG FIESTA, DOST Technology Transfer Day, etc. will also be encouraged. Potential technology adopters, as well as other collaborators and stakeholders, will be invited to attend the said activity. It is expected that the technologies developed would be able to reach the target adopters. Investment primers and flyers/leaflets of the commerciable technologies will be made to serve as promotional materials.

Also, in the second year, each RAISE will conduct an echo seminar/training to faculty researchers to further disseminate the learnings generated from this project. The IP-TBM staff will serve as resource persons to these echo-seminars.

8.2.5. Enhance Knowledge Management

IP Audit

The IP audit will be conducted by each participating agency to determine the types and number of IPs generated that can be included in IP protection and commercialization activities. Each participating agency is expected to submit an Inventory of IP assets and will be updated yearly to be assisted by the mentee agency.

Patent Mining

Patent mining will be done by the mentee agency and ParSU IP TBM staff will assist researchers know the technologies that should be develop concerning the selected commodity. The program will endorse AANR technologies for development as priority agenda for R&D. It will also conduct training on patent mining that the participants are the selected faculty and personnel of the mentor and mentee agencies.

Research Publication and Presentation to R&D For a Monitoring

Research to be published or for presentation to various fora will be requested to pass through RAISE office to be checked by IP TBM staff if there are possible significant disclosure of generated IP. This will be done to avoid public disclosure of IP's before its registration.

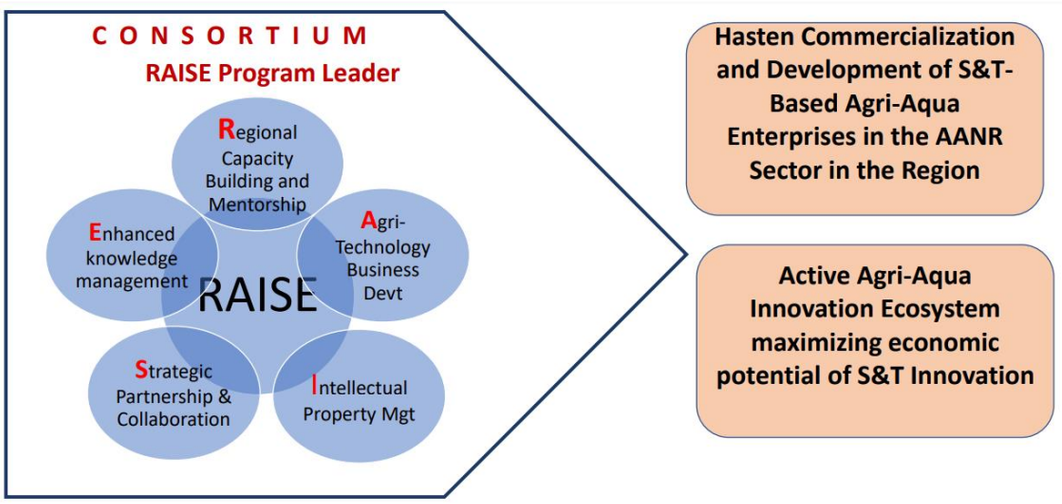
Research Inventory

The RAISE program will also be conducting an AANR research inventory every year. This Research inventory will be submitted to BCAARRD and PCAARRD by the ParSU IP TBM staff and will be encoded into the database of the RTMS.

Regional Technology Management System

A regional technology management system (RTMS) will be develop and will serve as a repository of different technologies from the mentor and mentee agencies. This website will be used by the faculty, students, and researchers as a source of related literature and studies and in patent searching. Though this RTMS there will be no duplication and re-inventing the wheel among the researches and technologies in the region and in the entire nation.

8.2.6 Conceptual Framework



Expected Outputs

Expected Output Project 1 (Old CMI)		
Y1 Output	**Y2 Output**	**Total**
Publications		
1 Training Module on IP Management 3 IEC	 2 IEC 1 Regional Sustainability Plan	1 Training Module on IP Management 5 IEC 1 Regional Sustainability Plan
Patents/ IP		
5 IP Applications	5 IP Applications	10 IP Applications
Product		
5 Prior art search of R&D Proposals 1 Institutional inventory of potential IPs 1 Institutional Inventory of IP Assets 1 Institutional Inventory of Mature Technologies 1 Product manufactured for pre-comm/ market tested 1 technology with pre-comm reports 1 Technology Commercialized	5 Prior art search of R&D Proposals 1 Institutional inventory of potential IPs 1 Institutional Inventory of IP Assets 1 Institutional Inventory of Mature Technologies 1 Product manufactured for pre-comm/ market tested 1 technology with pre-comm reports 1 Technology Commercialized	10 Prior art search of R&D Proposals 2 Institutional inventory of potential IPs 2 Institutional Inventory of IP Assets 2 Institutional Inventory of Mature Technologies 2 technology with pre-comm reports 2 Product manufactured for pre-comm/ market tested 2 Technology Commercialized
People and Services		
1 Institutional Prior Art Search & IP Audit Send at least 2 CMI Staff to IP Masterclass	At least 2 CMI Staff send to be trained for Technology Commercialization Mentorship Series 1 Institutional Pitch Day	1 Institutional Prior Art Search & IP Audit Send at least 2 CMI Staff to IP Masterclass At least 2 CMI Staff send to be trained for Technology Commercialization Mentorship Series 1 Institutional Pitch Day
Places and Partnerships		

1 Commercialization Agreement Signed	1 partnership agreement with Business Groups/Trade Institutions	1 Commercialization Agreement Signed 1 partnership agreement with Business Groups/Trade Institutions
Policy		

III. OTHER SUPPORTING DOCUMENTS REQUIRED (Please refer to page 2 for the additional necessary documents.)

DOST Form 3
NON-R&D PROJECT PROPOSAL

I. General Instruction: Submit through the DOST Project Management Information System (DPMIS), http://dpmis.dost.gov.ph, the non-R&D proposal and other applicable supporting documents required under item III below. Also, submit four (4) copies of the proposal together with its supporting documents. Use Arial font, 11 font size.

II. Operational Definition of Terms:

1. Title- the identification of the Program and the component projects.

Program- refers to a group of interrelated or complementing S&T projects that require an interdisciplinary or multidisciplinary approach to meet established goal(s) within a specific time frame.

Project- refers to the basic unit in the investigation of specific S&T problem/s with predetermined objective/s to be accomplished within a specific time frame.

2. Project Leader- refers to a project's principal researcher/implementer.

Agency- the institution of the Project Leader**.**

Smallest Unit of an Agency- refers to a Section, Division, Department or College provided that the Project Leader directly reports to the Head of said Section, Division, Department or College.

3. Cooperating Agency/ies- refers to the agency that support/s the project by participating in its implementation as collaborator, co-grantor, committed adopter of resulting technology, or potential investor in technology development or through other similar means.

4. Implementing Agency- the primary organization involved in the execution of a program/project which can be a public or private entity

Site/s of Implementation- location/s where the project will be conducted. Indicate the barangay, municipality, district, province, region, and country.

5. Project Duration- refers to the grant period or timeframe that covers the approved start and completion dates of the project, and the number of months the project will be implemented.

6. Project Cost- refers to the amount or budget requested by the Implementing/Monitoring Agency and/or approved by the Funding Agency.

Sustainable Development Goal (SDG) Addressed- indicates which among the 17 SDGs adopted by the United Nations Members States are addressed by the project

7. Executive Summary- briefly discusses what the whole proposal is about

8. Introduction- a formally written declaration of the project and its idea and context to explain the goals and objectives to be reached and other relevant information that explains the need for the project and aims to describe the amount of work planned for implementation; refers to a simple explanation or depiction of the project that can be used as communication material.

Rationale- brief analysis of the problems identified related to the project

Significance- refers to the alignment to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development.

Objectives- statements of the general and specific purposes to address the problem areas of the project.

Methodology- discusses the following: (a) variables or parameters to be measured and evaluated or analyzed; (b) treatments to be used and their layout; (c) experimental procedures and design; (d) statistical analysis; (e) evaluation method and observations to be made, strategies for implementation (Conceptual/Analytical framework).

Expected Outputs (6Ps)- deliverables of the project based on the 6Ps metrics (Publication, Patent/Intellectual Property, Product, People Service, Place and Partnership, and Policy).

Publication- published aspect of the research, or the whole of it, in a scientific journal or conference proceeding for peer review, or in a popular form.
Patent/Intellectual Property- proprietary invention or scientific process for potential future profit.
Product- invention with a potential for commercialization.
People Service- people or groups of people, who receive technical knowledge and training.
Place and Partnership- linkage forged because of the study.
Policy- science-based policy crafted and adopted by the government or academe as a result of the study.

Potential Outcomes- refer to the result that the proponent hopes to deliver three (3) years after the successful completion of the project.

Potential Impacts

Social Impact- refers to the effect or influence of the project to the reinforcement of social ties and building of local communities.
Economic Impact- refers to the effect or influence of the project to the commercialization of its products and services, improvement of the competitiveness of the private sector, and local, regional, and national economic development.

Target Beneficiaries- refers to groups/persons who will be positively affected by the conduct of the project.

Sustainability plan- refers to the continuity of the project or how it shall be operated amidst financial, social, and environmental risks.

Gender and Development (GAD) Score- refers to the result of accomplishing GAD checklists (for project monitoring and evaluation/project management and implementation) to highlight the contribution of the project in the achievement of the objectives of Republic Act 7192, "Women in Development and Nation Building Act," interpreted as gender-responsive, gender-sensitive, has promising GAD concepts, or GAD is invisible.

9. Workplan- refers to the plan of completing the project within a given time and in compliance to the set budget.

10. Project Management- includes discussion on the organizational set-up by which the project shall be implemented, and also on the monitoring scheme to be done by the Project Leader to ensure objectives are attained. Not to exceed one (1) page.

III. Other supporting documents required (as stated in Section VII of DOST A.O. No. 011, Series of 2020 – Revised Guidelines for the Grants-in-Aid Program):

1. Detailed breakdown of the required fund assistance to indicate the counterpart of the proponent and other fund sources including letter/s of commitment from the implementing, collaborating and coordinating agency/entity/ies;[1]
2. A counterpart fund, in kind and/or in cash, shall be required from the implementing agency/entity as one of the application requirements. All projects must have a minimum of 15% counterpart contribution except for projects involving public good;[1]
3. Curriculum Vitae or Personal Data Sheet (PDS) of Project Leader and other co-researchers/implementers. The service record may be requested if needed;[1]
4. Clearance from the DOST or the Funding Agency (e.g., DOST Councils) on previously funded completed projects handled by the Project Leader;[1]
5. Approval from the institution's ethics review board for research involving human subjects or in the case of animal subjects, approval from the Bureau of Animal Industry (BAI) (for PCAARRD- and PCHRD-monitored projects);
6. Clearance from the DOST Biosafety Committee (DOST-BC) shall be required for research proposals involving the use of GMOs under contained use (i.e., experiments done in laboratories, screen house, green house). For projects other than contained use, they shall be referred to the appropriate agency. The DOST Sectoral Councils, after determination as to whether or not the proposal has biosafety implications, shall endorse the same to the DOST-BC in accordance with the prescribed format under Annex 3 of the Philippine Biosafety Guidelines for Contained Use of Genetically Modified Organisms (series of 2014) (if applicable); and
7. For the private non-profit/non-government/people's organizations and startups:
    a. Up-to-date Securities and Exchange Commission (SEC) registration, or Department of Trade and Industry (DTI) registration, or Cooperative Development Authority

(CDA) registration certificate, or other authenticated copy of latest Articles of Cooperation and other related legal documents;
   b. Co-signers Statement (if applicable);
   c. Copy of latest Income Tax Return;
   d. Mayor's permit where the business is located;
   e. Audited Financial Statements for the past three (3) years preceding the date of project implementation or in case of those with operation of less than 3 years, for the years in operation and proof of previous implementation of similar projects (or in the case of startups, at least for one (1) year);
   f. Document showing that NGO/PO has equity to 20 percent of the total project cost, which shall be in the form of labor, land for the project site, facilities, equipment and the like, to be used in the project;
   g. Disclosure of other related business, if any;
   h. List and/or photographs of similar projects previously completed, if any, indicating the source of funds for implementation;
   i. Sworn affidavit of secretary of the NGO/PO that none of its incorporators, organizers, directors or officers is an agent of or related by consanguinity or affinity up to the fourth civil degree to the official of the agency authorized to process and/or approved the proposed MOA, and release of funds;
8. For CSOs, compliance to regulations as required by the General Appropriations Act (GAA) pertaining to fund transfers to Civil Society Organizations (CSOs); and
9. For foundations, DOST certification as accredited by the Science and Technology Foundation Unit

[1] *required of all proposals*

IV. Criteria for Evaluation:

A. Criteria for Evaluating Proposals

Criterion	Definition
Relevance or Significance	Aligned to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development
Technical / Scientific Merit	Sound scientific basis to generate new knowledge or apply existing knowledge in an innovative manner
Budget Appropriateness	The proposed budget is commensurate to the proposed work plan and deliverables.
Competence of Proponent	Proponent's expertise is relevant to the proposal and with proven competence to implement, manage and complete R&D programs/projects within the approved duration and budget.

B. Governing Council / Board and EXECOM's Evaluation Criteria

Criteria	Indicators	Raw Score
A. Soundness of Proposal (20%)	R&D addresses relevant sectoral need (applicable to pressing concern)	5
	Solution provided is most effective (compared to other proposed solutions)	5
	Proposed budget is reasonable (project is not expensive vis-a-vis output)	5
	Work plan is doable in a given timeframe	5
B. Suitability of Output	R&D output is cost-effective (cost is competitive in relation to new or existing products or process)	5

(30%)	Has identified partners to adopt the technology (with letter of support from the head of the company)	5
	Output can be commercialized (through an existing manufacturer, spin-off or start-up company)	5
	R&D utilization is timely (output should not be overtaken by other solutions)	5
C. Significance of Outcome (30%)	Economic: increase in productivity, increase in income, new jobs generated, high return of investment (ROI)	5
	Social: working partnerships established, training opportunities provided, policies adopted, increased access to basic services (i.e., food, health, education); political, cultural, gender sensitivity and inclusivity	5
	Environment: enhanced environmental health standards, no adverse effect to the environment	5
	Sustainability: sustainability mechanisms established in terms of institutional, financial and human resources capability (submission of a new proposal to sustain a completed or ongoing proposal does not constitute sustainability of the project)	5
D. Competence of Proponent (20%)	Proponent's expertise aligned with the proposal	5
	Collaboration with relevant agencies and/or industry partners	5
	Thorough understanding of the proposal's deliverables	5
	DOST has good experience with the proponent	5

C. Additional Criteria on Gender and Development (GAD)